



Geethanjali College of Engineering

Design and Analysis of Algorithms

Topic :Dijkstra's Algorithm

By

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What is Dijkstra's Algorithm?

- Proposed by **Edsger W. Dijkstra**
- Finds shortest paths from a source node
- Works on **non-negative weighted graphs**

Problem Statement

Input:

- Graph (V, E)
- Source node

Output:

- Shortest distance to all nodes

Core Idea

- Greedy approach
- Always pick the minimum distance unvisited node
- Relax edges (update distances)

💡 **Think:** expanding a ripple from the source 🌊

Algorithm Steps

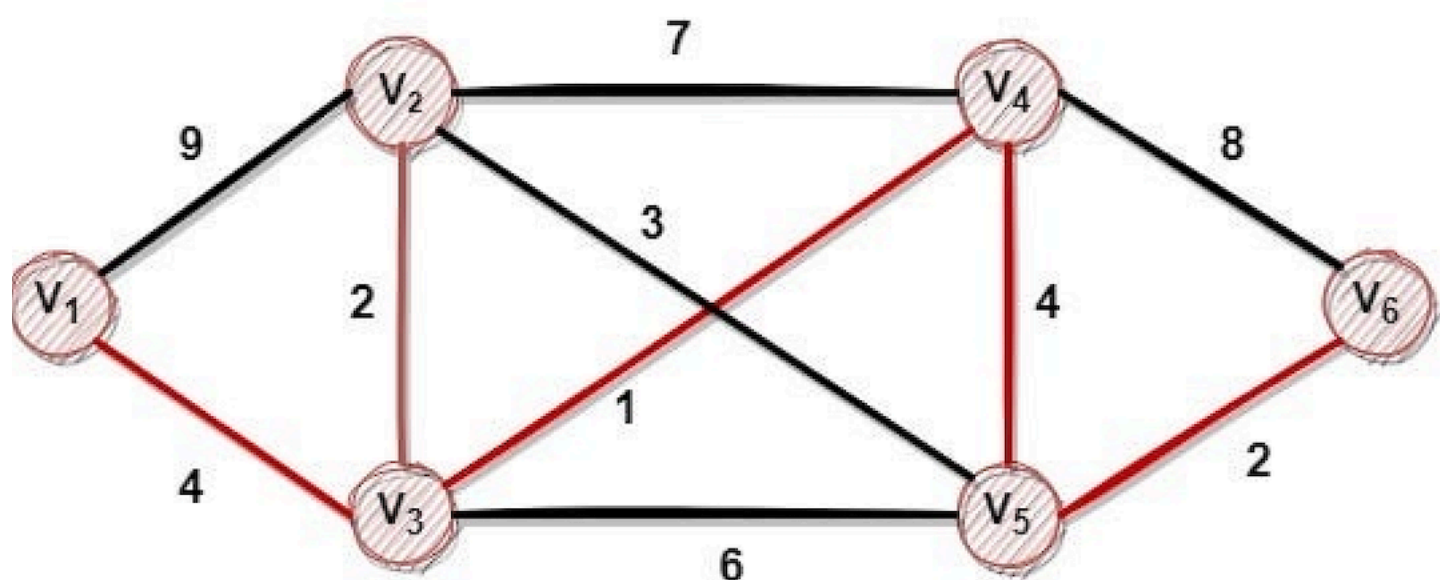
1. Initialize distances (∞ , source = 0)
2. Use priority queue
3. Extract min distance node
4. Relax neighbors
5. Repeat

Example Graph

Explanation:

The Path starts from vertex V1 and the next vertex is chosen by the minimal edge from the vertex V1 and the process continues till all the vertexes are visited

Dijkstra's Algorithm



Path: V₁ - V₃ - V₄ - V₅ - V₆

Total Cost: 11

	Distance	Previous Node	Visited
V ₁	0	None	True
V ₂	6	V ₃	True
V ₃	4	V ₁	True
V ₄	5	V ₃	True
V ₅	9	V ₄	True
V ₆	11	V ₅	True

Pseudocode

Initialize $\text{dist}[\text{source}] = 0$

All others $= \infty$

While PQ not empty:

u = node with min distance

for each neighbor v :

if $\text{dist}[u] + \text{weight} < \text{dist}[v]$:

 update $\text{dist}[v]$

Time Complexity

- Without heap $\rightarrow O(V^2)$
- With Min Heap $\rightarrow O((V + E) \log V)$

Edge Cases

1. Negative Edge Weights ❌

- Dijkstra fails
- Example: $A \rightarrow B$ (-5) breaks logic
👉 Use Bellman-Ford instead

2. Disconnected Graph ●

- Some nodes remain at ∞
- Means: no path exists

3. Multiple Shortest Paths 🔄

- Algorithm finds **one**, not all
- Depends on traversal order

4. Cycles in Graph 🔄

- Works fine
- But must avoid infinite loops using visited set

Limitations and Alternatives

Limitations:

- Cannot handle negative weights
- Memory heavy for dense graphs

Alternatives:

- Bellman-Ford $\rightarrow$ negative edges
- Floyd-Warshall $\rightarrow$ all pairs